Chapter 19: Digital Transformation

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 19 (pp. 757-779)

Overview

Chapter 19 (pp. 757-779) shows how the ADM structures a digital transformation: Phase A for the strategic vision, B for process redesign, C and D for the digital ecosystem and infrastructure, E-G for prioritizing, planning, and governing, and H for continuous innovation (p. 757). Section 19.1 defines digital transformation and its components (pp. 758-759). Section 19.2 relates it to Enterprise Architecture and distinguishes **digitization**, **digitalization**, and **digital transformation** (pp. 759-763). Section 19.3 gives nine ADM-based steps and a five-phase real hospital case (pp. 763-777). Section 19.4 summarizes (pp. 777-779). Leave with this idea: digital transformation starts with a change in the *business model*, not in IT. EA and transformation should be integrated in a hybrid approach (pp. 761-762, 769). Chapter 19 is part of the BLUE reading tour (p. 757).

> **Reading note:** The TOGAF view of digital transformation as the "strategic and systematic evolution of an organization's capabilities, processes, and technologies" is attributed to the Standard (p. 759). The nine steps are Jager's ADM-based approach. The hospital case (government policy "the right care in the right place", a 30% cost reduction) is his personal experience (pp. 768-777). Note that Ch. 3 (p. 48) points to "Chapter 18" for digital transformation, but it is covered here in Chapter 19.

Detailed Section Breakdown

Chapter Opener (p. 757)

- **ADM as the transformation vehicle.** A: strategic vision linking goals and digital capabilities. B: optimize processes for digital readiness. C and D: data, applications, and infrastructure for the digital ecosystem. E-G: continuous evaluation and governance. H: refine strategy as technology and needs evolve (p. 757).

19.1 What Is Digital Transformation? (pp. 758-759)

- **Definition.** Integrating digital technologies into all aspects of the organization to improve efficiency, customer experience, and innovation. A fundamental shift in how it operates and delivers value, "primarily based on a major change in the business and operating models" (p. 758).

Key component	Examples
Technology adoption	Cloud, AI, automation, analytics
Process optimization	Automated workflows, collaboration tools, AI insights
Customer experience enhancement	Digital platforms, self-service portals, personalization
Cultural change	Digital-first mindset, upskilling, agile
Data-driven decision-making	Big data, real-time analytics, BI

(pp. 758-759)

- **An ongoing journey**, not a one-time project (p. 759).
- **TOGAF's view.** "The strategic and systematic evolution of an organization's capabilities, processes, and technologies to enhance business outcomes through digital innovation", aligned with objectives,

governance, and efficiency (p. 759).

19.2 Digital Transformation and Enterprise Architecture (pp. 759-763)

- **Not just digitizing processes.** It requires a cultural shift. EA provides the framework that aligns technology, processes, and people with business objectives (p. 759).
- **Both are strategic, business-driven initiatives** that may require reorganizing internal and external information environments (p. 760).

Term	Meaning (Figure 19-1)
Digitization	Converting physical assets into digital formats (scanning, data conversion)
Digitalization	Using digital technology to modify existing processes and improve workflows
Digital transformation	Reimagining the business model and value propositions, from physical to predominantly digital engagement and delivery

(pp. 760-761)

- **Business-led.** "True transformation must come from within the business itself". Isolated IT or data strategies must align with business objectives. A strategy without the business structures is "a house without a solid foundation" (p. 761).
- **Mutual influence.** Transformation reshapes the Target Architecture and often starts with use cases scaled via the EA roadmap. EA provides the target, operating model, baseline, and gaps to set priorities and guide investment (p. 762).
- **Hybrid approach.** Integrating both gives immediate value plus a long-term foundation (p. 762).

19.3 A Structured Approach to Digital Transformation (pp. 763-777)

Step (Figure 19-2)	ADM phase	Key activities	Page
1. Prepare for transformation	Preliminary	Principles, governance, capabilities. Identify stakeholders (CxO, executives, architects). Set boundaries via transformation principles (customer centricity, data-driven, cloud-first). Digital maturity and readiness assessment. EA team and governance	p. 764
2. Define the digital strategy	A	Drivers (customer experience, efficiency, compliance) linked to SMART goals and KPIs. Enablers (AI, automation, cloud, IoT) in a high-level target. Stakeholder buy-in	pp. 764-765
3. Re-engineer business automation	B	Gap analysis of current vs. target business models. Redesign processes with automation. New roles. Customer-centric platforms. Business transformation roadmap (Work Package Portfolio Map)	p. 765
4. Design the digital ecosystem	C	Data: governance (GDPR, HIPAA), centralized strategy (data lakes, real-time analytics), AI insights. Applications: modernization (Application Portfolio Catalog), cloud and API integration	p. 766
5. Develop digital infrastructure	D	Cloud strategy (public, private, hybrid). Microservices and containers. Zero-trust and AI threat detection. High availability and DR. DevOps and CI/CD	p. 766
6. Create the roadmap	E	Prioritize by value and feasibility using the Business Value and Risk Matrix. Quick wins (high value, low risk). Short- and long-term initiatives	p. 767
7. Manage the implementation process	F	Incremental deployment (pilots, phased rollouts as Transitions). Risks and mitigation. Change management plan. Digital-first training	p. 767
8. Measure compliance and success	G	Governance frameworks, feedback loops, EA performance dashboards (Figure 12-54). Security, legal, and compliance risks	pp. 767-768

Step (Figure 19-2)	ADM phase	Key activities	Page
9. Ensure continuous improvement	H	Monitor trends, regular EA reviews, business-IT alignment, continuous innovation strategy	p. 768

Jager's hospital case (pp. 768-777)

- **Driver.** Government policy, "the right care in the right place", plus cost savings. The answer was home-based care, which required changing the business and operating model (telemedicine, remote monitoring, self-medication) (pp. 768-770).
- **Biggest challenge: people.** "Every change meets resistance". Many conversations were held to surface and address employee concerns (p. 769).

Case phase (Figure 19-3)	Months	ADM phases	Main activities
1. Assessment and strategy development	0-3	Preliminary, A	Assess IT (EHR, telemedicine), inefficiencies, a cybersecurity/HIPAA risk assessment, staff and patient feedback. Goals and KPIs (wait times, telehealth adoption). Board alignment (Table 19-1)
2. Technology selection and process optimization	3-6	B, C, D	Automate scheduling and billing, AI chatbots, digital HR, EHR enhancement, AI diagnostics, network security. New capabilities (Figure 19-4) (Table 19-2)
3. Implementation and staff training	6-12	E, F	Digital literacy and cybersecurity training, change management, telemedicine launch, wearables, AI clinical decision support (Table 19-3)
4. Patient experience and engagement	12-18	G	Patient portals, 24/7 AI virtual assistants, mobile apps, IoT smart beds, predictive analytics, RPA for claims (Table 19-4)
5. Continuous improvement and future innovations	18+	H	Track KPIs, gather feedback, update security, explore blockchain, AI drug discovery, AR/VR (Table 19-5)

(pp. 770-776)

- **Outcomes.** Virtual assistants handled non-acute consultations, freeing physicians for acute care (Figure 19-5). Healthcare costs were cut by 30% by moving non-acute care home. Staff came to see technology as an enabler (pp. 774-777).

19.4 Summary (pp. 777-779)

- Nine summarized steps: outline the digital vision; get ready for change; rethink business automation; build the digital framework; deploy the right technology; plan the journey; manage the transition; monitor progress and compliance; continue to improve. The ADM provides "a robust approach" for systematic transformation (pp. 777-779).

Key Terms & Definitions

Term	Definition	Page
Digital transformation	Integrating digital technology across the organization, based on a major change in business and operating models	p. 758
Digital transformation (TOGAF view)	Strategic, systematic evolution of capabilities, processes, and technologies to enhance outcomes through digital innovation	p. 759
Digitization	Converting physical assets to digital format	p. 760
Digitalization	Applying digital technology to modify existing processes	p. 760

Term	Definition	Page
Digital maturity assessment	Evaluation of customer engagement, workforce skills, processes, and infrastructure	p. 764
Hybrid approach	Integrating EA and digital transformation for immediate and long-term value	p. 762

Figures & Tables Referenced

Figure / Table	What it shows	Page
Figure 19-1	Focus areas of digitization, digitalization, digital transformation	p. 761
Figure 19-2	Nine steps for digital transformation using the ADM	p. 763
Figure 19-3	Five phases of the hospital transformation	p. 769
Tables 19-1 to 19-5	Activities per case phase	pp. 770-776
Figure 19-4	Introduction of new capabilities	p. 771
Figure 19-5	New capability removing the need for a physician role in non-acute cases	p. 774

Self-Check Questions

Q: Distinguish digitization, digitalization, and digital transformation.

A: Digitization converts physical assets to digital. Digitalization uses digital technology to improve existing processes. Digital transformation reimagines the business model and value propositions toward digital delivery (p. 760).

Q: Where must true digital transformation originate?

A: Within the business itself, as a change in the business model, not in IT (pp. 761, 769).

Q: How do EA and digital transformation influence each other?

A: Transformation insights reshape the Target Architecture and start from use cases scaled via the EA roadmap. EA supplies the target, operating model, baseline, and gaps for priorities and investment. A hybrid approach is best (p. 762).

Q: Which ADM phase defines the digital strategy, and what does it include?

A: Phase A: drivers, SMART goals and KPIs, enablers (AI, cloud, IoT) in a high-level target, and stakeholder buy-in (pp. 764-765).

Q: Which artifact helps identify quick-win transformation projects in Phase E?

A: The Business Value and Risk Matrix, which highlights high-value, low-risk projects (p. 767).

Q: Name the five key components of digital transformation.

A: Technology adoption, process optimization, customer experience enhancement, cultural change, data-driven decision-making (pp. 758-759).

Q: What was the biggest challenge in Jager's hospital case, and how was it handled?

A: Getting employees on board. Many conversations surfaced and addressed their concerns (p. 769).

Q: What result did the hospital achieve?

A: About 30% lower healthcare costs by moving non-acute care to home care through telehealth, remote monitoring, and self-medication (p. 777).